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EU AI Act

Post-Market Monitoring

Ongoing Oversight of AI Systems



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AI & Partners defends and extends the digital rights of users at risk around the world. By combining direct technical support, comprehensive policy engagement, global advocacy, grassroots professional services, regulatory interventions, and participating in industry groups such as AI Commons, we fight for fundamental rights in the artificial intelligence age.

This report was prepared by Sean Donald John Musch and Michael Charles Borrelli. For more information visit <https://www.ai-and-partners.com/>.

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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We’ve seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it’s handled responsibly. That’s why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

What Do We Do?

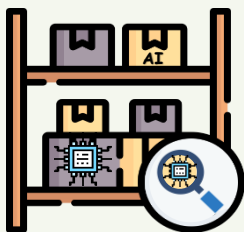
We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.



How Do We Do It?

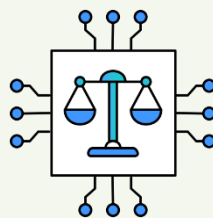
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building ‘Trustworthy-by-Design’.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.



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Introduction

As the EU Artificial Intelligence Act (EU AI Act) progresses toward full implementation, ensuring effective post-market monitoring of AI systems is a critical component for safeguarding public trust and promoting compliance. Harmonised standards, tailored for post-market monitoring, provide the necessary framework for continuous oversight. These standards help translate the legal obligations of the EU AI Act into actionable steps for firms and regulators, fostering innovation, trust, and accountability.

This report serves as a practical guide for stakeholders involved in AI post-market monitoring under the EU AI Act. It synthesizes best practices, highlights lessons learned from adjacent fields like cybersecurity, and offers concrete insights on fostering coherent, high-quality standards. From designing post-market monitoring systems to integrating them into existing compliance frameworks, the report details actionable steps for ensuring compliance, reducing operational risks, and fostering public trust.

By mapping the responsibilities and interactions among key actors, such as AI providers, regulators, and end users, this report promotes a collaborative, transparent, and inclusive approach. It aims to streamline AI compliance, support regulatory goals, and enhance the market's confidence in the safety and reliability of AI systems. Whether you're an AI provider ensuring the safety of your system, a regulator overseeing compliance, or an end user advocating for ethical AI, this document will serve as a valuable resource in navigating the evolving landscape of AI post-market monitoring under the EU AI Act.

Best regards,

Sean Musch

Founder/CEO

AI & Partners



The background is a dark blue to black gradient. It features a series of concentric circles that create a tunnel-like effect. The circles are composed of a grid of small squares, with the squares becoming more densely packed towards the center. A bright blue, glowing ring of light is positioned around the middle of the image, adding a sense of depth and focus.

Key questions being asked about Post-Market Monitoring



1. What is post-market monitoring in the context of the EU AI Act?

Post-market monitoring refers to the ongoing oversight of high-risk AI systems after they are placed on the market or put into service. It ensures the system continues to operate safely, lawfully, and in line with EU requirements throughout its lifecycle. The process involves collecting data, evaluating performance, detecting new risks, and documenting outcomes. It's not just a technical task—it's a legal obligation for providers of high-risk AI. The goal is to prevent harm, ensure accountability, and maintain public trust in AI technologies by identifying and mitigating risks that might not be evident pre-deployment.

2. Who is responsible for implementing post-market monitoring?

The primary responsibility lies with providers of high-risk AI systems. These are the entities that develop, market, or put the system into use under their name. Providers must design, document, and maintain a monitoring system proportionate to the risk and complexity of their AI solution. While deployers (i.e., users) may contribute data, the obligation to monitor and ensure continued compliance falls on the provider. In cases where AI systems are integrated into regulated sectors like finance or health, monitoring may also involve coordination with internal compliance teams or external regulators, but ultimate accountability remains with the provider.

3. What kind of data should be collected in post-market monitoring?

Relevant data includes technical performance metrics (accuracy, reliability, error rates), user feedback, incident reports, and evidence of system behavior in real-world conditions. Data should support evaluation of compliance with safety, transparency, and fairness standards. In some cases, interaction logs or feedback from deployers may be essential. However, providers must avoid collecting sensitive operational data from law enforcement agencies, respecting legal limits. Data should be anonymized and secured, especially if it contains personal or identifiable information. The aim is to detect performance drift, emerging risks, or any deviations from the AI system's intended and lawful use.

4. How does post-market monitoring relate to existing regulatory frameworks?

If an AI system falls under EU harmonisation legislation—such as regulations on medical devices or financial services—post-market monitoring can be integrated into existing compliance systems. This avoids duplication and reduces regulatory burden. However, the monitoring system must still fulfill all AI-specific requirements under Article 72, ensuring equivalent levels of risk control and accountability. Providers may use the EU Commission's forthcoming template to align both systems. Integration is especially useful in regulated industries, where providers already have risk management, quality control, and incident response systems in place that can be adapted to include AI oversight.

5. What is a post-market monitoring plan, and why is it important?

A post-market monitoring plan is a structured, pre-defined strategy detailing how a provider will monitor the AI system after launch. It is a mandatory part of the technical documentation and must include procedures for data collection, performance evaluation, and risk management. The EU Commission will issue a template to standardize these plans. The plan helps ensure that monitoring is not improvised or inconsistent. Instead, it reflects the AI system's risk profile and sector-specific context. This proactive approach enhances system reliability, simplifies audits, and reduces legal exposure by embedding long-term accountability from the very beginning.





6. What are the consequences of failing to comply with post-market monitoring obligations?

Non-compliance with Article 72 may result in serious consequences for providers, including administrative fines, withdrawal of the AI system from the market, or reputational damage. Regulators may impose penalties up to €30 million or 6% of global annual turnover, depending on the severity of the breach. Failing to monitor a high-risk system may also increase liability in case of harm to users. Moreover, inadequate post-market monitoring weakens public and stakeholder trust. To mitigate risk, providers must establish robust internal processes and keep detailed documentation demonstrating that monitoring was active, effective, and aligned with legal standards.

7. How often should post-market monitoring be conducted?

The EU AI Act doesn't specify an exact frequency, but monitoring should be ongoing, active, and proportionate to the system's risk level. This means that high-risk or mission-critical systems—like those used in healthcare, employment, or policing—may require continuous or real-time monitoring. In contrast, lower-risk high-risk systems may be reviewed periodically, such as quarterly or annually. Providers should also monitor following major updates, significant changes in deployment environments, or reported incidents. The post-market monitoring plan must define the appropriate frequency and triggers for evaluation based on the system's complexity, scale, and impact.

8. Are small or startup AI providers subject to the same requirements?

Yes, but the EU AI Act allows for proportionality based on the nature and risk of the system. Smaller providers or startups must still implement post-market monitoring if they develop high-risk AI systems, but the system can be scaled appropriately. For example, they may use third-party tools, open-source dashboards, or simplified data collection processes. However, they are still accountable for ensuring their system remains compliant and safe. Proportionality doesn't mean exemption—it means adapting the depth and structure of monitoring to fit the system's context while still meeting the core obligations of Article 72.

9. What should be included in the documentation of post-market monitoring?

Documentation must cover all aspects of monitoring: the post-market monitoring plan, data collection procedures, performance reviews, risk evaluations, incidents reported, corrective actions taken, and updates applied. Providers should also log stakeholder communications, audit results, and any regulatory reporting submitted. This documentation becomes part of the system's technical file and must be kept up to date and accessible for inspection by regulators. High-quality documentation is essential for traceability, internal learning, and legal defense. It should be comprehensive but also structured and searchable, making it easier to use for compliance audits and performance evaluations.

10. How does post-market monitoring support AI system improvement?

Monitoring provides a continuous feedback loop that reveals how the AI performs under real-world conditions. By analyzing collected data, providers can identify inefficiencies, user concerns, ethical risks, or technical failures. This insight allows them to refine algorithms, retrain models, improve user interfaces, and update documentation. It ensures the AI system evolves in line with user needs and legal obligations. Additionally, post-market data may inform future product development, helping providers build more robust and reliable AI. Thus, beyond compliance, post-market monitoring is also a strategic tool for innovation, quality assurance, and customer satisfaction.



The background features a series of concentric, glowing blue rings that create a tunnel-like effect. These rings are composed of a grid of small squares, with the inner rings being more densely packed and brighter. The overall color scheme is dark blue and black, with the glowing rings providing a strong contrast.

Understanding Post-Market Monitoring



1. System Design and Establishment



What does it involve?

This involves setting up a comprehensive system to monitor high-risk AI systems after they are placed on the market. The system must be proportionate to the risks and complexity of the AI product. It includes defining mechanisms for data collection, risk detection, and performance evaluation over the product's lifetime. Providers must integrate this system into their overall governance framework and technical documentation. The goal is to create a structured, proactive oversight mechanism that tracks the AI's real-world behavior and supports compliance with EU requirements across the entire operational lifecycle.

Why is it needed?

AI systems evolve and interact with dynamic real-world environments, which may lead to new risks after deployment. A post-market monitoring system helps ensure the AI remains compliant, safe, and effective throughout its use. It protects users, mitigates harm, and supports accountability. Without it, issues like algorithmic drift, bias, or unforeseen failures could go undetected, posing legal, ethical, and operational risks. This is especially vital for high-risk systems used in sensitive sectors like healthcare, employment, or law enforcement, where unchecked changes in system behavior could cause significant societal or individual harm.

How does it function?

The system operates by continuously or periodically collecting data from various sources, including end-users and system logs, to monitor real-world performance. Built-in feedback loops help detect anomalies, compliance gaps, or performance degradation. The design defines processes for evaluating this data and triggering appropriate responses, such as issuing updates or notifying authorities. Internal tools like dashboards, model tracking software, and alert systems may support the monitoring effort. All activities must be documented and reviewed regularly. The system also integrates with quality management systems and feeds into technical documentation to maintain a record of system oversight.





2. Planning (Pre-Market Stage)



What does it involve?

Before releasing a high-risk AI system, providers must create a post-market monitoring plan and include it in the technical documentation. This plan outlines how the provider will monitor the system after deployment—defining which data will be collected, how risks will be assessed, and what actions will be taken. The European Commission will provide a standardized template by February 2026 to guide structure and content. The plan is foundational to operationalizing the monitoring system and must reflect the AI system's risk profile and operating environment, including sector-specific considerations.

Why is it needed?

A pre-market plan ensures that providers think ahead about post-deployment risks rather than reacting to them after incidents occur. It institutionalizes long-term accountability and embeds compliance into the AI system's lifecycle. This proactive planning reduces the likelihood of harm, helps anticipate technical or ethical issues, and supports a consistent approach to managing emerging challenges. The requirement also ensures that providers don't overlook monitoring responsibilities in the rush to market. For regulators, it provides a concrete, auditable framework, increasing transparency and setting expectations for post-market behavior.

How does it function?

The plan functions as a blueprint for post-market activities. It defines the processes, data sources, performance metrics, responsible roles, and reporting channels for monitoring the AI system. Providers must implement these steps in practice once the system is deployed. The plan also becomes a reference point for audits or regulatory review. If problems arise, the plan should guide investigations and remediation. It is a living document that may evolve over time as providers gain more insights from real-world performance. Alignment with the Commission's forthcoming template ensures harmonization across different providers and use cases.





3. Data Collection (Ongoing)



What does it involve?

Providers must continuously collect and document relevant performance data on their AI systems. This data may come from system logs, user feedback, operational environments, or integrated sensors. The goal is to capture how the AI behaves in real-world use, identifying deviations from expected performance, emerging risks, or technical failures. Data collection must respect privacy laws and avoid sensitive operational data in law enforcement contexts. The collected data forms the foundation for evaluating whether the AI system continues to meet legal, ethical, and safety standards under the EU AI Act.

Why is it needed?

AI systems often change once deployed due to updates, learning mechanisms, or environmental variation. Without continuous data, providers can't detect issues like model drift, bias, or misuse. Regular data collection allows for timely corrections, minimizing harm to users and avoiding regulatory violations. It also demonstrates that providers are actively engaged in maintaining compliance, not just at the design stage but throughout the product's lifecycle. For regulators and clients, ongoing data provides essential evidence of responsible governance. Ultimately, it helps ensure that AI systems behave consistently with their intended, certified functions.

How does it function?

Providers implement tools and processes—such as APIs, user reporting interfaces, log analyzers, and telemetry systems—to collect operational data. This may involve collaboration with clients or deployers to access real-time system performance data. The data is then stored securely, anonymized as needed, and fed into analytics systems or dashboards for review. Alerts or threshold checks may flag concerning trends or anomalies. This triggers further evaluation or corrective action. The system must document all data collection activities and outcomes, ensuring traceability and accountability during audits or investigations.





4. Performance Monitoring and Risk Evaluation



What does it involve?

This stage involves analyzing the collected data to evaluate whether the AI system continues to perform correctly and safely. Providers must check for degradation, errors, bias, or unintended behavior in the system's outputs. It also includes risk assessments based on real-world interactions, evolving use cases, or changes in data inputs. Providers must compare current performance against compliance criteria defined in Chapter III, Section 2 of the EU AI Act. This ongoing evaluation ensures the system meets its obligations and flags when updates, interventions, or even system withdrawal are necessary.

Why is it needed?

Even if an AI system meets regulatory standards at launch, real-world deployment can introduce unexpected variables. New risks may emerge due to environmental changes, user behavior, or input data shifts. Performance monitoring identifies these early, reducing harm and liability. It also supports continuous compliance with EU safety and fairness requirements. Without regular evaluation, issues could go unnoticed until significant damage occurs—undermining public trust and prompting costly enforcement. This stage ensures that AI systems adapt responsibly to evolving conditions and that providers stay accountable for system behavior throughout its operational life.

How does it function?

Performance monitoring uses predefined metrics and evaluation criteria to assess system outputs. Automated tools can track accuracy, fairness, latency, and other indicators. Risk evaluation involves reviewing flagged anomalies, assessing impact severity, and updating risk registers. Providers may use internal reviews, expert panels, or even third-party audits to support this process. If problems are detected, the provider must document the findings and take corrective action. Monitoring results are often fed back into the technical documentation and may trigger updates to the system itself or the monitoring plan. The cycle repeats regularly to ensure continuity.





5. Integration with Existing Regulatory System



What does it involve?

If a provider already operates under existing EU harmonisation legislation—such as those governing medical devices or financial systems—they may integrate AI post-market monitoring into those existing compliance systems. This approach avoids duplicating oversight mechanisms. However, the integrated system must still meet the same protective standards required under the EU AI Act. Providers must use the Commission’s post-market monitoring template where appropriate and adapt existing documentation, controls, and data pipelines to ensure full coverage of AI-specific requirements while leveraging their current governance and quality systems.

Why is it needed?

This integration reduces administrative burden, promotes regulatory coherence, and streamlines compliance for businesses already subject to EU sector-specific laws. It recognizes the complexity of regulated industries where layered compliance frameworks already exist. By allowing integration, the EU avoids overregulating and duplicating effort—especially in finance, healthcare, and energy. It also helps regulators cross-reference data and oversight across domains. For providers, it creates efficiencies and reduces the cost of compliance. For society, it ensures AI risks are managed without gaps or overlaps in regulatory protections, even when multiple compliance regimes intersect.

How does it function?

Providers map AI-specific monitoring requirements onto existing governance processes—like risk assessments, quality management systems, or audit trails. They adapt their systems to incorporate AI performance metrics and documentation protocols. The post-market monitoring template from the EU Commission helps harmonize this integration. Oversight teams ensure that these merged systems continue to meet all AI Act obligations, especially around transparency, accountability, and risk mitigation. Internal or external compliance audits may verify equivalence. This approach is particularly useful in institutions with mature regulatory operations, allowing AI governance to scale within already-functioning systems while maintaining protective standards.





6. Reporting and Documentation



What does it involve?

This final stage involves systematically documenting all post-market monitoring activities and results. Providers must keep records of data collected, performance assessments, risk evaluations, and corrective actions taken. The documentation becomes part of the technical file required under Annex IV of the EU AI Act. It must be clear, complete, and ready for inspection by regulators. Providers must also report certain incidents or risks to authorities, especially if user safety or legal compliance is affected. This stage formalizes transparency, helps demonstrate due diligence, and supports traceability throughout the AI system's operational life.

Why is it needed?

Thorough documentation ensures that all monitoring and decision-making processes are traceable, verifiable, and accountable. It protects providers by showing that reasonable steps were taken to ensure safety and compliance. It also supports regulators in assessing system behavior, identifying trends, and enforcing the law. If incidents arise, documented evidence can clarify the provider's response and guide improvements. In highly regulated sectors, it helps clients and auditors understand AI system governance. Ultimately, documentation underpins the credibility of post-market monitoring—without it, oversight loses structure, and stakeholders lose confidence in providers' responsibility.

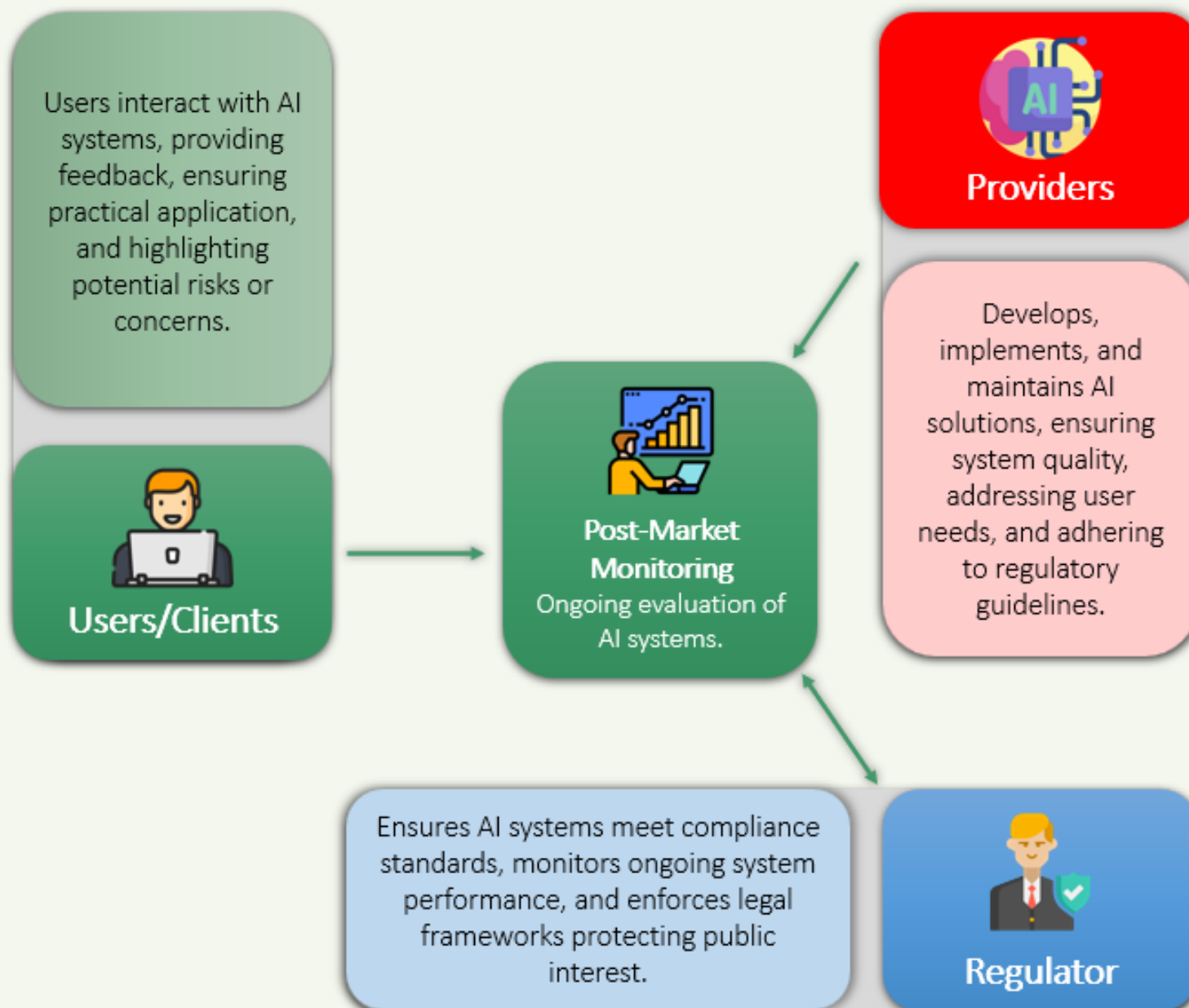
How does it function?

Providers maintain logs, reports, dashboards, and incident records—ideally in a centralized system linked to the technical documentation. They must ensure consistency, accuracy, and accessibility of this documentation for regulatory audits. Regular reviews and updates may be scheduled to maintain compliance. If a risk or anomaly is found, the system logs the detection, analysis, response, and resolution. In serious cases, mandatory reporting to EU authorities may be triggered. Documentation also feeds into internal review cycles and strategic planning, ensuring the AI lifecycle remains compliant, safe, and adaptable as conditions change.



The background features a series of concentric circles composed of small, dark blue squares, creating a tunnel-like effect. A bright, glowing blue ring is positioned in the center, illuminating the surrounding pattern.

Mapping Post-Market Monitoring to Ecosystem Actors





Post-Market Monitoring		Actors		
Area	Description	Users/Clients	Providers	Regulators
System Design and Establishment	Providers must create a post-market monitoring system aligned with the AI system's complexity and risk. This foundational step ensures the system is capable of identifying compliance issues over time. It must be proportionate, scalable, and integrated into the provider's broader AI governance and quality management processes.	System design ensures that the AI products clients interact with are built with long-term oversight in mind. This benefits users by reducing risks associated with malfunctioning, biased, or unpredictable behavior over time. It also demonstrates that the provider has taken proactive steps to prevent harm and ensure reliability. Especially in high-risk applications like healthcare or finance, users will have increased trust in solutions that are continuously monitored. While they might not interact directly with the monitoring system, its existence provides an assurance that the AI will remain compliant, safe, and performant during its full operational lifecycle.	Providers bear the primary responsibility for establishing a post-market monitoring system. They must assess risks, select appropriate tools and frameworks, and document procedures. This requires cross-functional input—engineering, legal, compliance, and product management all contribute. While it imposes upfront resource demands, the benefit is long-term risk mitigation and streamlined future compliance. Designing a proportionate monitoring system also offers scalability and cost efficiency. For innovative or rapidly evolving products, a well-designed system can also drive improvements and updates. However, failure to design effectively could result in regulatory sanctions, market access barriers, or reputational harm if non-compliance is discovered post-deployment.	Regulators benefit from a structured and mandatory system design requirement because it sets a minimum baseline for all providers. This facilitates enforcement, especially for high-risk AI systems, where public trust is vital. Knowing that a monitoring system must exist and is tailored to the AI's risk profile allows regulators to focus resources on evaluating the effectiveness rather than debating its necessity. It also encourages industry-wide consistency and helps streamline audit and inspection practices. Moreover, requiring this from the outset increases legal clarity, reducing ambiguity in what's expected from companies across sectors and sizes, supporting fair and transparent oversight.
Planning (Pre-Market Stage)	A post-market monitoring plan must be developed before market placement and included in the	Having a defined post-market monitoring plan before deployment reassures clients that safety and	Providers are required to craft a detailed monitoring plan and integrate it into technical	A pre-market monitoring plan gives regulators a concrete document to evaluate during compliance



	technical documentation. The plan outlines procedures for data collection, analysis, and compliance evaluation. The European Commission will provide a standardized template to guide content and structure, ensuring consistency and completeness across all providers.	compliance are not afterthoughts. It provides confidence that their AI providers have mapped out how they will identify and handle problems over time. This is especially important for clients in regulated industries, where demonstrating oversight and governance is part of their own compliance obligations. A pre-market plan also signals that the provider takes long-term accountability seriously, which may influence vendor selection. While users may never see the full plan, knowing one exists—and is based on an EU-approved template—adds a layer of credibility to the system.	documentation. This involves not just setting intentions but defining procedures for data collection, analysis, risk evaluation, and mitigation. The plan must be structured, comprehensive, and aligned with the Commission's upcoming template. For providers, this may require legal and regulatory interpretation, collaboration across teams, and potential updates to internal processes. While it may initially add a layer of bureaucracy, it ultimately serves to protect against future liabilities and enforcement actions. It also lays the groundwork for continuous improvement and better communication with clients and authorities.	reviews. It ensures that providers are thinking ahead and integrating monitoring into their lifecycle strategy. The Commission's template will harmonize expectations across sectors and countries, allowing regulators to more easily assess whether a provider's plan meets minimum requirements. It also offers a clear audit trail—plans can be compared against actual monitoring results later. For new technologies or novel applications, this plan acts as a vital point of reference to understand how the provider intends to safeguard public interest and manage emerging risks post-launch.
Data Collection (Ongoing)	Monitoring systems must actively collect and document relevant data throughout the AI system's lifecycle. This includes feedback from deployers and technical performance data from operational environments. Data collection must respect legal	Ongoing data collection helps ensure that the AI system continues to operate reliably, ethically, and safely after it's been deployed. Clients benefit from adaptive systems that respond to real-world usage and flag issues early. This is particularly	Providers must develop robust infrastructure and processes to collect and store relevant operational data, sometimes in real-time. This includes ensuring GDPR compliance, protecting personal and sensitive data, and maintaining cybersecurity standards. They	Ongoing data collection gives regulators insight into how AI systems behave in practice, beyond lab conditions. It enables more proactive supervision and allows them to intervene if patterns of non-compliance or harm emerge. The





	boundaries, especially regarding sensitive information, and must be continuous, systematic, and sufficient to evaluate compliance.	important in applications where the AI learns or updates over time, as real-world drift could introduce new risks. Clients may be asked to share certain performance metrics or feedback, which could require coordination. As long as privacy and data protection are respected, users generally benefit from systems that are actively monitored—issues can be detected and resolved before causing significant harm.	must determine what types of data are relevant for monitoring and build tools or APIs to collect it. This may also involve partnerships with clients or third parties to access performance data. Done well, data collection supports both compliance and business intelligence. Done poorly, it could introduce legal, reputational, or operational risks. Providers must also ensure they are not collecting sensitive law enforcement data in prohibited ways.	quality and scope of collected data also reflect a provider's commitment to responsible AI governance. Regulators will likely request or inspect such data during audits or investigations. Having real-time or periodic data enables better policymaking and risk forecasting across the sector. Additionally, it helps close the gap between theoretical compliance and real-world outcomes, offering a more accurate picture of system impact on end-users.
Performance Monitoring and Risk Evaluation	Collected data must be analyzed to monitor the AI system's performance, detect risks, and ensure compliance with regulatory requirements. Evaluation should also consider how the system interacts with other AI systems. This stage is essential to identify non-conformities early and support proactive risk management and system updates.	Clients benefit significantly from continuous performance monitoring, as it ensures the AI system remains safe, accurate, and relevant throughout its use. Issues such as bias, misclassification, or model drift can be detected early, avoiding negative outcomes or harm. For mission-critical applications, this monitoring boosts reliability and trust. Users may also receive updates or reports based on these evaluations, giving them insights into system quality.	Providers must interpret collected data to assess the AI system's performance and compliance with legal and ethical requirements. This includes evaluating output accuracy, system reliability, and potential risks like bias or drift. The process may involve deploying dashboards, anomaly detection tools, or human oversight. Risk evaluation becomes part of the feedback loop, leading to system improvements or patches. Providers must also document these	For regulators, performance monitoring is critical in ensuring that high-risk AI systems not only start compliant but stay compliant. It enables authorities to assess whether systems operate safely in changing environments. Evaluation data allows regulators to track patterns, compare across providers, and spot systemic risks. It supports enforcement by providing a basis for requiring corrections or even market withdrawal. Regulators may also issue new





		Furthermore, by addressing risks promptly, providers can avoid disruptions in service that might otherwise affect clients' operations, reputation, or compliance standing—especially in healthcare, finance, or infrastructure environments.	evaluations and any corrective actions taken. While this is resource-intensive, it helps reduce liability, enables continuous improvement, and can serve as proof of due diligence in case of regulatory inquiry or legal challenge.	guidance based on emerging trends identified through this process. Overall, this stage strengthens the lifecycle accountability model and enables more dynamic regulatory oversight based on how systems behave in real-world conditions.
Integration with Existing Regulatory System	If AI systems are already subject to sectoral regulations, such as those in finance or medical devices, providers can integrate post-market monitoring into existing frameworks. This avoids duplication, streamlines compliance, and ensures consistency—as long as the integrated system maintains an equivalent level of legal protection and oversight.	For users in highly regulated sectors—such as finance, healthcare, or transportation—integrating AI post-market monitoring with existing oversight frameworks reduces compliance complexity. It ensures that AI safety doesn't conflict with other legal or operational obligations. Clients benefit from more coherent governance, smoother vendor relationships, and fewer redundant audits or documentation requests. It also increases confidence that the AI system complies with both sectoral regulations and AI-specific obligations. This dual assurance becomes especially important when AI decisions directly impact customer wellbeing, financial	Providers operating in sectors already governed by EU harmonisation legislation can avoid duplicative efforts by aligning AI monitoring with their existing regulatory systems. This enables a more efficient use of resources and simplifies internal governance structures. However, providers must ensure that integration still meets the EU AI Act's level of protection—merely “bolting on” AI monitoring won't suffice. Legal and technical teams must collaborate to verify equivalency. For financial institutions or medical device manufacturers, this alignment can also offer a competitive edge by demonstrating regulatory agility and holistic risk management capabilities to	Regulators benefit from smoother coordination between AI oversight and existing sectoral regulators, such as financial or health authorities. Integration supports consistent risk management standards across regulatory domains without adding burdensome duplication. It allows for cross-checking compliance across frameworks, supporting a “regulate once, comply many times” approach. However, regulators must ensure that the integrated systems do not water down AI-specific obligations. Oversight may involve cooperative arrangements between different regulatory bodies. Ultimately, this integration fosters efficiency while





		outcomes, or legal accountability.	clients and regulators alike.	reinforcing protection standards, ensuring that AI risks are managed in tandem with other sector-specific risks without creating silos.
Reporting and Documentation	All monitoring results, assessments, and corrective actions must be documented. This ensures traceability, supports accountability, and facilitates audits or regulatory	For clients, robust documentation practices provide assurance that the AI system they are using is under continuous and responsible oversight. They may have access to performance summaries, incident reports, or system updates, depending on contractual agreements. This transparency helps clients meet their own compliance obligations and manage third-party risk. If something goes wrong, clear documentation supports faster investigation and resolution. In regulated industries, such records may also be necessary for audits or supervisory reviews. Over time, documented monitoring supports more informed procurement and risk decisions, contributing to a more mature and trustworthy AI market.	Providers are required to meticulously document monitoring activities, performance evaluations, risk assessments, and any corrective measures taken. This documentation forms the backbone of regulatory compliance and is vital during audits, legal disputes, or public inquiries. It also contributes to internal accountability, ensuring issues are tracked and addressed systematically. While documentation can be time-consuming, it protects the provider by showing evidence of responsible governance and due diligence. Moreover, good documentation facilitates internal learning, helping teams refine models, spot recurring issues, and improve future system development.	For regulators, documentation is essential to effective enforcement and policy evolution. It allows them to verify that monitoring systems are in place and functioning as claimed. Documentation supports audits, complaint investigations, and post-incident inquiries, providing a paper trail of how risks were identified and managed. Regulators can also analyze aggregated documentation across sectors to detect trends, benchmark performance, or revise regulatory guidance. If documentation is missing, inconsistent, or clearly fabricated, it signals potential non-compliance and may justify enforcement action. Ultimately, it's the backbone of transparency and accountability in AI governance under the EU AI Act.



The background is a complex, abstract geometric pattern in shades of blue and black. It features concentric circles and rings of small squares and lines, creating a tunnel-like or digital effect. Two prominent, glowing blue rings are visible, adding a sense of depth and energy to the design.

Calls to Action



Build a Scalable, Risk-Based Monitoring Architecture

Design a post-market monitoring (PMM) system aligned with the AI system's complexity, use case, and risk category. Ensure it is modular, scalable, and integrated with broader AI governance and quality management systems. Embed risk scoring mechanisms, anomaly detection tools, and compliance checkpoints from the outset. The system should evolve as the AI model is updated or retrained, ensuring continuous alignment with legal, ethical, and operational standards



Develop a Pre-Launch Monitoring Plan with Clear Roles and Metrics

Before releasing the AI system, draft a comprehensive PMM plan and embed it into technical documentation. Define performance indicators, risk thresholds, data sources, responsibilities, and reporting workflows. Align the plan with EU regulatory expectations and forthcoming Commission templates. Cross-functional teams—legal, compliance, & tech — should co-develop the plan to ensure it reflects real-world challenges and oversight needs.



Set Up a Continuous Feedback and Logging Infrastructure

Implement systems to collect and log user feedback, error reports, system behavior metrics, and environmental interactions throughout the AI system's lifecycle. Integrate dashboards and alerts for real-time visibility. Ensure data privacy and security by applying access controls, anonymization where needed, and compliance with GDPR and other data protection regulations. This data forms the backbone for future compliance assessments and model updates



Conduct Routine Risk Audits and Cross- System Performance Reviews

Regularly analyze operational data to detect emerging risks, assess regulatory alignment, and evaluate AI interactions with other systems. Use KPIs, red-flag triggers, and trend analysis to identify anomalies or declining performance. Schedule formal evaluations at set intervals and in response to significant updates. Engage compliance and technical experts to review results and recommend changes proactively.





Conclusion

The introduction of harmonised standards under the EU Artificial Intelligence Act (EU AI Act) represents a significant shift in how AI systems are developed, assessed, and deployed. This move promotes a more consistent, transparent, and legally sound approach to compliance. By providing clear technical specifications and unified benchmarks, harmonised standards help transform the legal obligations of the AI Act into practical, actionable steps, reducing uncertainties for innovators and supporting regulatory oversight, while safeguarding fundamental rights.

This report serves as a strategic guide for stakeholders engaged in the AI standard-setting process, offering insights into how harmonised standards can streamline compliance and foster innovation across the EU. It highlights the role of these standards in aligning with key EU AI Act requirements, such as data governance, human oversight, transparency, and lifecycle risk management, all while promoting trust and market confidence.

However, the successful adoption of harmonised standards requires more than just technical specifications. It necessitates operational readiness, proactive engagement, and continuous alignment with evolving legal and societal expectations.

Stakeholders—including AI firms, regulators, and civil society actors—must adapt their compliance systems, product development processes, and governance frameworks to these new standards.

For smaller companies and first-time participants, early engagement with standardisation bodies and collaborative advocacy is critical to overcoming challenges and ensuring a smooth integration into the regulatory landscape.

Despite these complexities, forward-thinking organisations and regulators are already demonstrating that harmonised standards are essential in driving not only compliance and product quality but also fostering market confidence. As AI technologies continue to evolve and expand across various sectors—from healthcare and financial services to manufacturing and mobility—harmonised standards will be key in ensuring that AI innovation remains technically robust, legally compliant, and ethically aligned with European values.





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